Stage V - Technology Hub Evaluation System Processes	
Version	2.1
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Organization:	Truth Collective / ISI Consulting
Classification:	Public
Document Type:	Stage V - Technology Hub Evaluation System Processes
Date:	Aug-26

Purpose of the Technology Hub Evaluation System

The Technology Hub Evaluation System governs all engineered, digital, computational, and electronic products featured within the Technology Hub. This includes:

- Computers
- Laptops
- Tablets
- Smartphones
- Networking devices
- Smart home controllers
- Digital interfaces
- Audio/visual electronics
- Peripheral devices
- Storage hardware
- Processing hardware

This governance document ensures that every technology product meets professional engineering standards, compliance requirements, and category-specific performance thresholds before being recommended.

1. Governing Bodies & Standards Authorities

Technology products must align with the following internationally recognized engineering and compliance bodies:

Electrical & Computing Standards

- **IEEE** — Institute of Electrical and Electronics Engineers
- **IEC** — International Electrotechnical Commission
- **ISO/IEC 27001** — Information security
- **ISO/IEC 62368** — Audio/Video & ICT equipment safety

Semiconductor & Memory Standards

- **JEDEC** — Joint Electron Device Engineering Council
- **PCI-SIG** — PCI Express standards
- **USB-IF** — USB Implementers Forum

Safety & Compliance

- **UL** — Electrical safety
- **ETL** — Intertek certification
- **CSA Group** — Safety compliance
- **FCC** — Electromagnetic emissions
- **CE** — European conformity
- **RoHS** — Hazardous substance restrictions

Products must meet all applicable compliance requirements to be eligible for recommendation.

2. Mission-Critical Component Requirements

Technology products are evaluated based on mission-critical components that materially determine performance, reliability, and longevity.

Computing Devices (Laptops, Desktops, Tablets)

- CPU architecture
- GPU capability
- Thermal management
- PCB layer count
- Power delivery stability
- SSD controller type
- RAM timing and bandwidth
- Chassis thermal dissipation
- Battery safety and lifecycle

Networking Devices

- Protocol reliability
- Throughput performance
- Antenna design
- Firmware stability
- Security compliance
- Thermal performance

Smart Home Controllers

- Sensor accuracy
- Automation protocol reliability

- Firmware integrity
- Interoperability standards
- Electrical safety

Audio/Visual Electronics

- Driver quality
- Signal processing accuracy
- DAC/ADC performance
- Display panel quality
- Color accuracy
- Refresh stability

Mission-critical components carry the highest weight in the scoring model.

3. Weighted Engineering Scoring Model

Technology products are scored using Truth Collective's weighted engineering model:

Component Tier	Weight	Engineering Impact
Mission-Critical	50–60%	Determines core performance, reliability, and safety
Important	25–35%	Influences consistency and operational stability
Minor	10–15%	Affects usability and secondary experience

Weights vary by product type based on governing bodies and engineering relevance.

4. Category-Specific Performance Metrics

Each technology product category has its own performance benchmarks.

Computing Performance Metrics

- Thermal throttling behavior
- Sustained performance under load
- Power efficiency
- Memory bandwidth
- Storage throughput
- GPU rendering stability

Networking Performance Metrics

- Latency
- Packet stability
- Throughput

- Range consistency
- Security protocol compliance

Smart Home Performance Metrics

- Sensor precision
- Automation reliability
- Interoperability
- Firmware update stability

Audio/Visual Performance Metrics

- Frequency response accuracy
- Total harmonic distortion
- Display color accuracy
- Refresh rate stability
- Signal integrity

Products must meet or exceed category thresholds to be recommended.

5. Material & Construction Requirements

Technology products are evaluated for:

Material Quality

- Chassis material grade
- Heat sink composition
- PCB quality
- Connector durability
- Insulation quality

Construction Integrity

- Internal bracing
- Component mounting quality
- Thermal pathway design
- Fastener quality
- Assembly precision

Poor construction integrity results in reduced scoring.

6. Reliability & Lifecycle Requirements

Technology products must demonstrate:

- Thermal stability
- Firmware reliability
- Component fatigue resistance
- Battery lifecycle durability
- Long-term performance consistency
- Manufacturer support
- Warranty strength

Products with reliability concerns are downgraded or disqualified.

7. Compliance Verification

All technology products must provide verifiable compliance evidence:

- UL / ETL / CSA safety
- FCC emissions
- CE conformity
- RoHS material safety
- ISO/IEC category standards

Missing compliance results in automatic disqualification.

8. Truth Collective Tier Assignment

Technology products are assigned a tier based on Stage I and Stage II results:

Tier 1 — Premium Engineering Grade

Meets or exceeds all major engineering benchmarks and compliance requirements.

Tier 2 — Professional Grade

Meets most benchmarks with minor limitations.

Tier 3 — Consumer Grade

Meets minimum standards for general use.

Tier 4 — Entry Level

Fails multiple benchmarks or lacks compliance.

Tiering is based solely on engineering evidence.

9. Evidence Requirements

Accepted evidence includes:

- Manufacturer engineering documentation
- Benchmark results
- Teardown analysis
- Compliance certificates
- Reviewer observations
- Material specifications

Marketing claims and consumer star ratings are excluded.

10. Technology Hub Output

Each evaluation produces:

- Standards mapping
- Mission-critical component analysis
- Weighted engineering score
- Category performance score
- Reliability score
- Compliance verification
- Final tier assignment
- Recommendation rationale

This ensures all Technology Hub recommendations are grounded in measurable engineering evidence.

Conclusion

The Technology Hub Evaluation System ensures that every technology product featured by Truth Collective meets professional engineering standards, compliance requirements, and category-specific performance thresholds. This governance framework maintains accuracy, integrity, and trust across all technology recommendations.